

Changing hostname:

```
j@Javeria-F2023266353:~$ hostnamectl
hostname
  Static hostname: Javeria-F2023266353
        Icon name: computer-vm
        Chassis: vm 🖥️
        Machine ID: d2eebc6959f74e14ae9491174ce37487
        Boot ID: 15eee683cd394e26bb651dd21e983d7c
  Virtualization: qemu
  Operating System: Ubuntu 25.10
        Kernel: Linux 6.17.0-35-generic
  Architecture: arm64
  Hardware Vendor: QEMU
  Hardware Model: QEMU Virtual Machine
  Firmware Version: 0.0.0
        Firmware Date: Fri 2015-02-06
        Firmware Age: 11y 4month 1w 6d
Javeria-F2023266353
```

Step 1 — Update System:

```
j@j-QEMU-Virtual-Machine:~$ sudo nano /etc/hosts
j@j-QEMU-Virtual-Machine:~$ sudo apt update
sudo apt upgrade -y
Hit:1 http://ports.ubuntu.com/ubuntu-ports questing InRelease
Hit:2 http://ports.ubuntu.com/ubuntu-ports questing-updates InRelease
Hit:3 http://ports.ubuntu.com/ubuntu-ports questing-backports InRelease
Hit:4 http://ports.ubuntu.com/ubuntu-ports questing-security InRelease
78 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  linux-headers-6.17.0-5      linux-tools-6.17.0-5
  linux-headers-6.17.0-5-generic linux-tools-6.17.0-5-generic
  linux-modules-6.17.0-5-generic
Use 'sudo apt autoremove' to remove them.

Upgrading:
  alsa-ucm-conf      libgphoto2-port12t64
  apparmor           libinput-bin
  bluez              libinput10
  bluez-cups         libinsights0
  bluez-obexd        libldap-common
  cloud-init         libldap2
  cloud-init-base    libnft-common
  crash              libnft-core
```

Step 2: Install Java

Step 3: Install OpenSSH

```
Created symlink '/etc/systemd/system/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-06-21 10:35:42 PKT; 6ms ago
 Invocation: d66bb1efe29a4098a9aafd0c83a867d7
 TriggeredBy: ● ssh.socket
               Docs: man:ssh(8)
                  man:ssh_config(5)
   Process: 20399 ExecStartPre=/usr/sbin/ssh -t (code=exited, status=0/SUCCESS)
  Main PID: 20401 (sshd)
    Tasks: 1 (limit: 3942)
   Memory: 1.5M (peak: 2M)
      CPU: 11ms
   CGroup: /system.slice/ssh.service
           └─20401 "sshd: /usr/sbin/ssh -D [listener] 0 of 10-100 startups"

Jun 21 10:35:42 Javeria-F2023266353 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server:
Jun 21 10:35:42 Javeria-F2023266353 sshd[20401]: Server listening on 0.0.0.0 port 22.
Jun 21 10:35:42 Javeria-F2023266353 sshd[20401]: Server listening on :: port 22.
Jun 21 10:35:42 Javeria-F2023266353 systemd[1]: Started ssh.service - OpenBSD Secure Shell server:
lines 1-19/19 (END)
```

Step 4: Create Hadoop Group

```
@Javeria-F2023266353:~$ sudo groupadd hadoop
[sudo: authenticate] Password:
@Javeria-F2023266353:~$ getent group hadoop
hadoop:x:1001:
@Javeria-F2023266353:~$
```

Step 5: Create Hadoop User

```
j@Javeria-F2023266353:~$ sudo groupadd hadoop
[sudo: authenticate] Password:
j@Javeria-F2023266353:~$ getent group hadoop
hadoop:x:1001:
j@Javeria-F2023266353:~$ sudo useradd -m -g hadoop -s /bin/bash hadoop
j@Javeria-F2023266353:~$ sudo passwd hadoop
New password:
```

Step 6: Assign Permissions

```
j@Javeria-F2023266353:~$ sudo mkdir -p /usr/local/hadoop
j@Javeria-F2023266353:~$ sudo chown -R hadoop:hadoop /usr/local/hadoop
j@Javeria-F2023266353:~$ ls -ld /usr/local/hadoop
drwxr-xr-x 2 hadoop hadoop 4096 Jun 21 18:43 /usr/local/hadoop
j@Javeria-F2023266353:~$
```

Step 7: Generate SSH Keys and Configure Passwordless Login

```
hadoop@Javeria-F2023266353:~$ ssh-keygen -t rsa -P ""
Generating public/private rsa key pair.
Enter file in which to save the key (/home/hadoop/.ssh/id_rsa):
Created directory '/home/hadoop/.ssh'.
Your identification has been saved in /home/hadoop/.ssh/id_rsa
Your public key has been saved in /home/hadoop/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:UfoGd4Gz5rZcZhk0LKssb6SamhZwgYEOkFvLPk2SyPo hadoop@Javeria-F2023266353
The key's randomart image is:
+---[RSA 3072]-----+
|o+          ...    |
|* .          o+ .   |
|+. .        +o.+ .  |
|o..         .-*.    |
|. + o .     S=o.    |
| .+ + o ..+ +      |
| ... B . o *       |
|.o . = =   o       |
|+oEo.=.           |
+---[SHA256]-----+
hadoop@Javeria-F2023266353:~$ mkdir -p ~/.ssh
chmod 700 ~/.ssh
```

Step 8: Download and Install Hadoop 3.4.1

```
hadoop@Javeria-F2023266353:~$ cd -  
tar -xzf hadoop-3.4.1.tar.gz  
hadoop@Javeria-F2023266353:~$ ls  
hadoop-3.4.1  hadoop-3.4.1.tar.gz  snap  
hadoop@Javeria-F2023266353:~$
```

```
hadoop@Javeria-F2023266353:~$ ls /home  
hadoop j  
hadoop@Javeria-F2023266353:~$ su - j  
Password:  
j@Javeria-F2023266353:~$ whoami  
j  
j@Javeria-F2023266353:~$ sudo chown -R hadoop:hadoop /usr/local/hadoop  
[sudo: authenticate] Password:  
j@Javeria-F2023266353:~$ ls -ld /usr/local/hadoop  
drwxr-xr-x 2 hadoop hadoop 4096 Jun 21 10:43 /usr/local/hadoop  
j@Javeria-F2023266353:~$ su - hadoop  
Password:  
hadoop@Javeria-F2023266353:~$ whoami  
hadoop  
hadoop@Javeria-F2023266353:~$
```

Step 9: Move Hadoop to Installation Directory

```
hadoop@Javeria-F2023266353:~$ nano ~/.bashrc
hadoop@Javeria-F2023266353:~$ source ~/.bashrc
hadoop@Javeria-F2023266353:~$ echo $JAVA_HOME
echo $HADOOP_HOME
/usr/lib/jvm/java-11-openjdk-arm64
/usr/local/hadoop
hadoop@Javeria-F2023266353:~$
```

```
GNU nano 8.4 /home/hadoop/.bashrc
# ~/.bashrc: executed by bash(1) for non-login shells.
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
# for examples

# If not running interactively, don't do anything
case $- in
  *i*) ;;
  *) return;;
esac

# don't put duplicate lines or lines starting with space in the history.
[ Read 135 lines ]

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^I Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Videos

Downloads

Step 10: Configure hadoop-env.sh

```
hadoop@Javeria-F2023266353:~$ su - hadoop
Password:
hadoop@Javeria-F2023266353:~$ nano ~/.bashrc
hadoop@Javeria-F2023266353:~$ echo $JAVA_HOME
echo $HADOOP_HOME
/usr/lib/jvm/java-11-openjdk-arm64
/usr/local/hadoop
hadoop@Javeria-F2023266353:~$ source ~/.bashrc
hadoop@Javeria-F2023266353:~$ echo $JAVA_HOME
echo $HADOOP_HOME
/usr/lib/jvm/java-11-openjdk-arm64
/usr/local/hadoop
hadoop@Javeria-F2023266353:~$
```

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-arm64
export HADOOP_HOME=/usr/local/hadoop
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
```

Help Write Out Where Is Cut Execute Location
Exit Read File Replace Paste Justify Go To Line

Step 11: Configure core-site.xml

```
hadoop@Javeria-F2023266353: ~ — ssh: x j@Javeria-F2023266353: ~
GNU nano 8.4 /usr/local/hadoop/etc/hadoop/core-site.xml
<configuration>

  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>

</configuration>

[ Read 8 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
Videos
Downloads
```

```
hadoop@Javeria-F2023266353: ~ — ssh: x j@Javeria-F2023266353: ~
GNU nano 8.4 /usr/local/hadoop/etc/hadoop/hdfs-site.xml *
<configuration>

  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>

  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///home/hadoop/hdfs/namenode</value>
  </property>

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
Videos
```



```
hadoop@Javerla-F2023266353:~$ cat /usr/local/hadoop/etc/hadoop/hdfs-site.xml
<configuration>

  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>

  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///home/hadoop/hdfs/namenode</value>
  </property>

  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///home/hadoop/hdfs/datanode</value>
  </property>

</configuration>
```

Step 12: Configure hdfs-site.xml

```
GNU nano 8.4 /usr/local/hadoop/etc/hadoop/hdfs-site.xml *
<configuration>

  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>

  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:///home/hadoop/hdfs/namenode</value>
  </property>

  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:///home/hadoop/hdfs/datanode</value>
  </property>

</configuration>

^H Help      ^O Write Out ^F Where Is  ^K Cut       ^J Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^L Paste     ^_ Justify   ^_ Go To Line
```

```
hadoop@Javeria-F2023266353:~$ nano /usr/local/hadoop/etc/hadoop/hdfs-site.xml
hadoop@Javeria-F2023266353:~$ mkdir -p ~/hdfs/namenode
mkdir -p ~/hdfs/datanode
hadoop@Javeria-F2023266353:~$ ls -R ~/hdfs
/home/hadoop/hdfs:
datanode  namenode

/home/hadoop/hdfs/datanode:

/home/hadoop/hdfs/namenode:
hadoop@Javeria-F2023266353:~$
```

Step 13: Configure mapred-site.xml

```
GNU nano 8.4 /usr/local/hadoop/etc/hadoop/mapred-site.xml
<configuration>

  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>

</configuration>

hadoop@Javeria-F2023266353:~$ cp /usr/local/hadoop/etc/hadoop/mapred-site.xml.te
mplate \
/usr/local/hadoop/etc/hadoop/mapred-site.xml
cp: cannot stat '/usr/local/hadoop/etc/hadoop/mapred-site.xml.template': No such
file or directory
hadoop@Javeria-F2023266353:~$ nano /usr/local/hadoop/etc/hadoop/mapred-site.xml
```

Step 14: Configure yarn-site.xml

```
hadoop@Javeria-F2023266353: ~$ nano /usr/local/hadoop/etc/hadoop/yarn-site.xml
GNU nano 8.4 /usr/local/hadoop/etc/hadoop/yarn-site.xml #
<configuration>

  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>

</configuration>
```

```
hadoop@Javeria-F2023266353:~$ nano /usr/local/hadoop/etc/hadoop/yarn-site.xml
hadoop@Javeria-F2023266353:~$ cat /usr/local/hadoop/etc/hadoop/mapred-site.xml
<configuration>

  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>

</configuration>
hadoop@Javeria-F2023266353:~$ cat /usr/local/hadoop/etc/hadoop/yarn-site.xml
<configuration>

  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>

</configuration>
hadoop@Javeria-F2023266353:~$
```

Step 15: Format the HDFS NameNode

```
hadoop@Javeria-F2023266353:~$ hdfs namenode -format
2026-06-21 21:44:54,173 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = Javeria-F2023266353/127.0.1.1
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.4.1
STARTUP_MSG: classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/
hadoop/common/lib/netty-transport-native-unix-common-4.1.100.Final.jar:/usr/local/
hadoop/share/hadoop/common/lib/netty-buffer-4.1.100.Final.jar:/usr/local/hadoop
p/share/hadoop/common/lib/jersey-server-1.19.4.jar:/usr/local/hadoop/share/hadoo
p/common/lib/netty-all-4.1.100.Final.jar:/usr/local/hadoop/share/hadoop/common/l
ib/curator-client-5.2.0.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-co
llections-3.2.2.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty-http-9.4.53.
v20231009.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-classes-
kqueue-4.1.100.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/slf4j-reload4
j-1.7.36.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-udt-4.1.1
00.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-transport-classes-e
poll-4.1.100.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/jzdbc-annotati
ons-1.1.jar:/usr/local/hadoop/share/hadoop/common/lib/javax.servlet-api-3.1.0.jar
:/usr/local/hadoop/share/hadoop/common/lib/kerby-asn1-2.0.3.jar:/usr/local/hadoo
p/share/hadoop/common/lib/kerby-config-2.0.3.jar:/usr/local/hadoop/share/hadoo
p/common/lib/commons-math3-3.6.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jc
tty-xml-9.4.53.v20231009.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-n
et-3.9.0.jar:/usr/local/hadoop/share/hadoop/common/lib/jersey-servlet-1.19.4.jar
```

Step 16: Start Hadoop Services

```
ved in 0 seconds .
2026-06-21 21:45:09,424 INFO namenode.NYSStorageRetentionManager: Going to retain
1 images with txid >= 0
2026-06-21 21:45:09,426 INFO blockmanagement.DatanodeManager: Slow peers collect
ion thread shutdown
2026-06-21 21:45:09,432 INFO namenode.FSNamesystem: Stopping services started fo
r active state
2026-06-21 21:45:09,432 INFO namenode.FSNamesystem: Stopping services started fo
r standby state
2026-06-21 21:45:09,434 INFO namenode.FSImage: FSImageSaver clean checkpoint: tx
id=0 when next shutdown.
2026-06-21 21:45:09,434 INFO namenode.NameNode: SHUTDOWN_MSG: I
/*****
SHUTDOWN_MSG: Shutting down NameNode at Javeria-F2023266353/127.0.1.1
*****/
hadoop@Javeria-F2023266353:~$ start-dfs.sh
Starting namenodes on [localhost]
localhost: ERROR: JAVA_HOME is not set and could not be found.
Starting datanodes
localhost: ERROR: JAVA_HOME is not set and could not be found.
Starting secondary namenodes [Javeria-F2023266353]
Javeria-F2023266353: Warning: Permanently added 'Javeria-F2023266353' (ED25519)
to the list of known hosts.
Javeria-F2023266353: ERROR: JAVA_HOME is not set and could not be found.
2026-06-21 21:46:39,541 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
```

Step 17: Verify Running Hadoop Services

```
hadoop@Javeria-F2023266353:~$ start-yarn.sh
Starting resourcemanager
resourcemanager is running as process 29472. stop it first and ensure /tmp/hado
op-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanagers
hadoop@Javeria-F2023266353:~$ jps
38800 NameNode
29472 ResourceManager
31602 Jps
31204 SecondaryNameNode
31001 DataNode
31517 NodeManager
hadoop@Javeria-F2023266353:~$
```

Step 18 : Download/Create WordCount Input File

```
hadoop@Javeria-F2023266353:~$ ls -lh ~/assignment
total 72M
-rw-r--r-- 1 hadoop hadoop 72M Jun 21 22:17 hdp.txt
-rw-rw-r-- 1 hadoop hadoop  0 Jun 21 22:01 input.txt
hadoop@Javeria-F2023266353:~$
```

Step 19 :Create Assignment Folder & Place Input File

```
hadoop@Javeria-F2023266353:~$ ls ~/assignment
hdp.txt  input.txt
hadoop@Javeria-F2023266353:~$ pwd
/home/hadoop
```

Step 20 : Upload Input File to HDFS

```
/home/hadoop
hadoop@Javeria-F2023266353:~$ hdfs dfs -ls /input
2026-06-21 22:30:44,160 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r-- 1 hadoop supergroup 75380115 2026-06-21 22:18 /input/hdp.txt
hadoop@Javeria-F2023266353:~$
```

Step 21 : Run WordCount

```
hadoop@Javeria-F2023266353:~$ hdfs dfs -ls /output
2026-06-21 22:32:47,318 WARN util.NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 hadoop supergroup 0 2026-06-21 22:27 /output/_SUCCESS
-rw-r--r-- 1 hadoop supergroup 8099884 2026-06-21 22:27 /output/part-r-00000
```


Step 22 : View Output

```
hadoop@Javerla-F2023266353:~$ hdfs dfs -cat /output/part-r-00000 | head -20
2026-06-21 22:33:43,665 WARN util.NativeCodeLoader: Unable to load native-hadoop
  library for your platform... using builtin-java classes where applicable
#      4
(each  1
0      216
1      16
10     24
1000   2
10000  16
100000 2
100001 1
100002 4
100003 3
100004 1
100005 2
100006 1
100007 1
100008 7
100009 15
10001  2
100010 1
100012 1
```